

Product Carbon Footprint Study Report

Product Name: passenger car production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: passenger car production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg passenger car, petrol/natural gas

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg passenger car, petrol/natural gas.

2. System Boundary

passenger car production, petrol/natural gas, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of passenger car production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **3.487 kg CO2-Eq**.

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

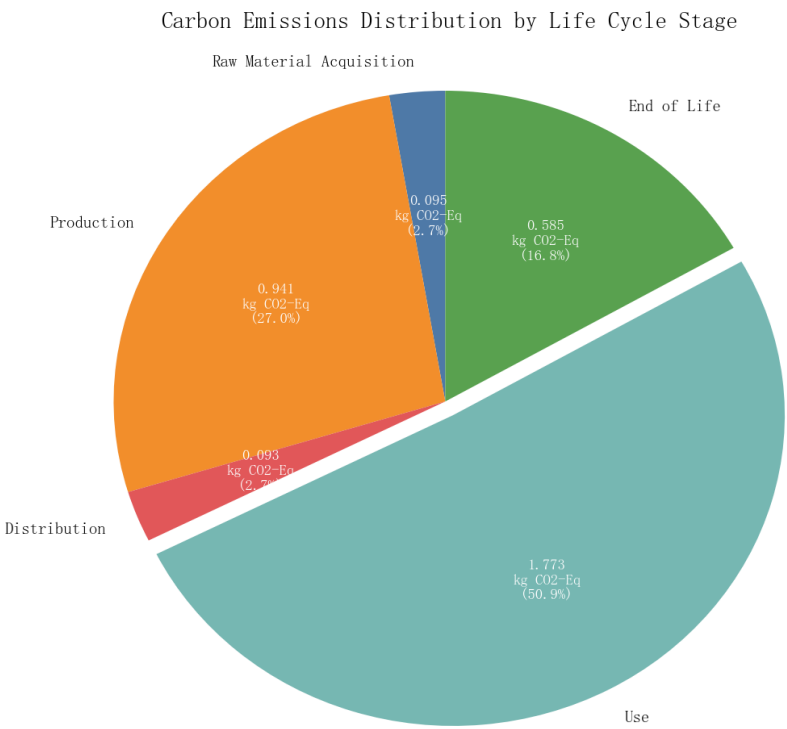


Table 1: Carbon Emissions by Life Cycle Stage

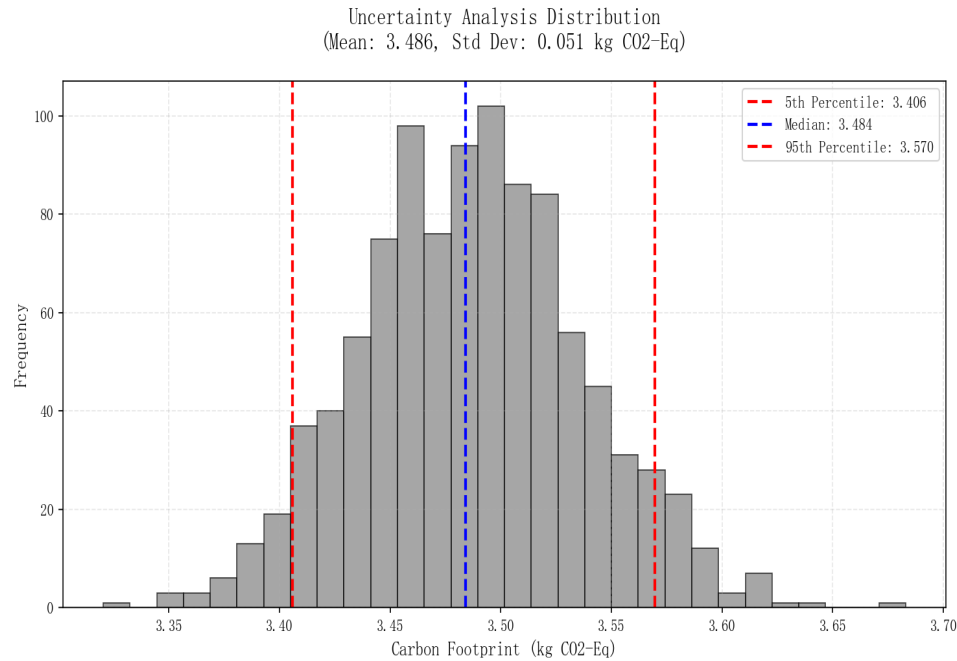
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.095	2.71
Production	0.941	27.00
Distribution	0.093	2.67

Use	1.773	50.85
End of Life	0.585	16.77
Total	3.487	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	3.486
Standard Deviation	0.051
Coefficient of Variation (CV)	1.47%
Median	3.484
5th Percentile	3.406
95th Percentile	3.570

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of passenger car production stands at 3.487 kg CO₂-Eq. This figure represents a compact yet comprehensive cradle-to-grave assessment, highlighting the product's environmental intensity under the defined functional unit. While modest in absolute terms, the result provides a clear benchmark for comparing alternative designs or material substitutions within the automotive sector.

Lifecycle contributions reveal distinct patterns: the use phase dominates at 50.9% (1.773 kg CO₂-Eq), followed by production at 27.0% (0.941 kg CO₂-Eq) and end-of-life at 16.8% (0.585 kg CO₂-Eq). Raw material extraction and distribution each contribute only 2.7%, indicating that operational energy demand and manufacturing processes constitute the primary emission drivers. These proportions underscore the need to prioritize efficiency improvements in vehicle operation and factory energy use.

Major emission sources are therefore concentrated in the use and production stages, which together account for nearly 78% of the total impact. Targeted interventions here offer the greatest leverage for overall footprint reduction.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the selection of a representative European electricity mix for the use phase, average transport distances for distribution, and standard recycling rates at end-of-life. The functional unit is defined as one passenger car over its typical service life, with allocation procedures applied to multi-output processes according to ISO 14044 guidelines.

Limitations arise from reliance on secondary data sources such as Ecoinvent and industry averages, which may not fully capture site-specific variations in energy efficiency or supplier performance. Data quality for certain raw materials remains moderate due to temporal and geographical mismatches, potentially introducing uncertainty in the lower-contribution stages.

System boundary restrictions encompass a cradle-to-grave scope that excludes infrastructure-related impacts, indirect land-use changes, and downstream consumer behavior beyond the defined use phase. Capital goods and maintenance activities outside core manufacturing were also omitted to maintain focus on direct product-related emissions.

3. Suggestions for Improvement

- Target the use phase (50.9%) by integrating lightweight materials and powertrain efficiency upgrades to lower operational energy demand.
- Optimize production-stage emissions (27.0%) through renewable electricity procurement and process heat recovery systems in assembly plants.
- Enhance end-of-life recovery (16.8%) via design-for-disassembly protocols and partnerships with advanced recycling facilities to increase material circularity.

- Establish primary data collection from key suppliers to improve accuracy of raw-material and production inventories.
- Conduct sensitivity analyses on electricity grid scenarios to identify robust low-carbon pathways across different regional markets.